Fermentation-derived / single-cell protein ingredients (precision-fermented, cultured, yeast-biomass proteins): HMTc Standard

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CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

The Heavy Metal Tested & Certified human-grade standard for precision-fermented, cultured, and single-cell / yeast-biomass protein ingredients used in pet food, stated as sold on a dry powder basis. Pet food is legally feed; HMTc voluntarily certifies each analyte to the strictest human-food sovereign maximum for the nearest analog, with the laxer feed-law floor disclosed for contrast.

1 | How every number on this standard is set

This subcategory covers a precision-fermented / single-cell / cultured PROTEIN INGREDIENT (spray-dried, heat-inactivated microbial biomass, e.g. Saccharomyces cerevisiae expressing an animal-identical protein) sold as a constituent of pet food. It is legally FEED (FDA CVM review, AAFCO, EU Directive 2002/32). Two read-across facts set the number honestly. First, a single-cell-protein biomass is not contamination-free by construction: anionic cell-wall groups biosorb Pb/Cd/Ni/Cr, arsenate is actively imported through the yeast phosphate transporters, and the metal load tracks the fermentation feedstock (plant sugars/molasses) and the mineral/trace-element premix — so Cd, Pb, and inorganic arsenic are real, load-bearing risks and are never marked not-material. Second, the strictest applicable HUMAN-FOOD analog on a dry as-sold basis is the plant-protein fraction (cereals, pulses, legumes), the same strict analog set the finished pet food already reads across; adopting it (rather than the looser same-class spec) keeps the ingredient limit at least as strict as any finished food it enters (Method v2.0 strictest-applicable rule, and the anti-circumvention guardrail against

a looser ingredient tier laundering into a stricter finished food). The resulting toxic ceilings (Pb 20, Cd 40, tAs 100, iAs 100, tHg 20 ug/kg) are stricter on every toxic than the only governed same-class human-food spec, the EU novel-food Yarrowia lipolytica yeast-biomass Union-list contaminant limits (Pb 100, Cd 100, As 150, Hg 100 ug/kg; Reg (EU) 2019/760 amending 2017/2470), which is disclosed as the corroborating over-compliance floor and never selects the number. The applicable feed floor is disclosed on every row and never selects the number.

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	HMTC LIMIT	BASIS
Lead (Pb)	20	read-across (Hong Kong)
Cadmium (Cd)	40	read-across (EU 2023/915)
Total arsenic (tAs)	100	read-across (Paraguay)
Inorganic arsenic (iAs)	100	read-across (Israel)
Mercury, total (tHg)	20	read-across (Hong Kong)
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	800	read-across (EU 2024/1987)
Tin (Sn)	250,000	read-across (Singapore)
Aluminium (Al)	100,000	read-across (China)
Total chromium (Cr)	500	read-across (Hong Kong)
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	20	read-across (Hong Kong); Cereal-based food intended to be consumed principally by persons under the age of 36 months
Cadmium (Cd)	40	read-across (EU 2023/915); pulses, general

Total arsenic (tAs)	100	read-across (Paraguay); pulses, dried legume seeds, except soybean
Inorganic arsenic (iAs)	100	read-across (Israel); Dried legumes, including peanuts and soybeans
Mercury, total (tHg)	20	read-across (Hong Kong); Rice, husked rice, polished rice, maize, maize flour, wheat, wheat flour
Methylmercury (MeHg)	≤ tHg	bounded by tHg (reflex speciation)
Nickel (Ni)	800	read-across (EU 2024/1987); cereals except durum/rice/husked-rice/-pseudocereal/oat rows
Tin (Sn)	250,000	read-across (Singapore); any food
Aluminium (Al)	100,000	read-across (China); baked foods (bread, pastries, cakes, biscuits, cookies, crackers); deep-fried flour products; batter, breading, and frying powder
Total chromium (Cr)	500	read-across (Hong Kong); Vegetables, other than pulses
Hexavalent chromium (Cr-VI)	≤ Cr	bounded by Cr (reflex speciation)

4 | Cite this standard

Pendergrass, K. (2026). *Fermentation-derived / single-cell protein ingredients (precision-fermented, cultured, yeast-biomass proteins): HMTc standard* (Standard No. 10.83138/hmtc.std.17.pet-food-fermentation-derived-protein). Institute of Contaminant Standards. https://doi.org/10.83138/hmtc.std.17.pet-food-fermentation-derived-protein

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References

[1] Hong Kong
Binds Pb, tHg, Cr.
READ-ACROSS

[2] EU 2023/915
Binds Cd.
READ-ACROSS

[3] Paraguay
Binds tAs.
READ-ACROSS

[4] EU 2024/1987
Binds Ni.
READ-ACROSS

[5] Singapore
Binds Sn. any food.
READ-ACROSS

[6] China
Binds Al. baked foods (bread, pastries, cakes, biscuits, cookies, crackers); deep-fried flour products; batter, breading, and frying powder.
READ-ACROSS

[7] Israel
Binds iAs.
READ-ACROSS

STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

COMPETING INTERESTS

This standard is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

DATA AVAILABILITY

The works this standard draws on are listed in full in the references above and on the standard page at https://heavymetalcertified.com/standards/pet-food-fermentation-derived-protein. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/pet-food-fermentation-derived-protein.